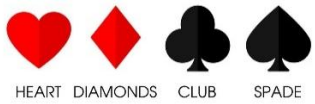
COM410 Programming in Practice

Assignment 2: Individual Development Exercise (60%)

Submission Deadline: 12pm (Noon), 04th May 2026

This assignment carries 60% of the overall mark for the module
This assignment addresses module Learning Outcomes 3 and 4



High Card Series

Introduction

High Card Series is a card game designed for between **two and five players** and is played using a standard shuffled deck of playing cards. At the start of the game, the number of rounds to be played is specified, with the total number of rounds restricted to a minimum of **five** and a maximum of **ten**. In each round, every player is dealt **a single card** from the deck. The cards dealt in that round are then compared based on their rank, with card ranks ordered from highest to lowest as **Ace, King, Queen, Jack**, followed by the numerical cards from **ten down to two**.

The player who receives the highest-ranked card in a given round keeps the card and is awarded **three points**. In cases where two or more players receive cards of equal highest rank, the round is considered tied between those players. In such situations, each tied player keeps their card and is awarded **one point**, with no additional tie-breaking applied. Cards dealt to players who do not achieve the highest rank in the round are **returned to the deck and placed at the bottom of the remaining cards**. This process is repeated for each round of the game, allowing the deck to be reused throughout while players accumulate both **points** and a **collection of cards** obtained during successful or tied rounds.

Once all rounds have been completed, each player is given a single optional opportunity to adjust their collected cards before bonus points are calculated. During this adjustment stage, a player may choose to discard **one or two cards** from their collection and draw the same number of replacement cards from the deck. Replacement cards are drawn from the top of the deck. Each player may use this option **at most once**, and players may also choose not to make any changes.

Following the completion of the optional adjustment stage, **bonus points** are awarded based on an evaluation of the cards collected by each player over the course of the game. Each player first sorts their collected cards by rank in order to identify the **longest sequence of consecutive card values** they possess. The player who achieves the longest such sequence is awarded **five bonus points**; in the event of a tie for the longest sequence, each tied player is awarded **two bonus points**.

After this, each player counts the number of cards they have collected for each suit. For every player, the **highest count achieved in any single suit** is identified, and the player whose suit count is highest among all players is awarded **five bonus points**. If two or more players are tied for the highest suit count, **each tied player receives two bonus points**.

The game concludes once all rounds have been completed and bonus points have been applied. Each player's final score is calculated by combining points earned during the rounds with any bonus points awarded at the end. The player with the highest final score is declared the winner; if final scores are equal, the game is declared a draw.

Example Scenario (Three Players, 6 Rounds)

For example, a **three-player game consisting of 6 rounds** might proceed as follows.

Round-by-Round Play

Round	P1 Card	P2 Card	P3 Card	Round Winner	Cards Kept	Points Awarded
1	9♦	K♣	7♥	Player 2	Player 2 keeps K♣	P2 +3
2	A♠	A♦	10♣	Tie (P1, P2)	P1 keeps A♠, P2 keeps A♦	P1 +1, P2 +1
3	Q♥	J♠	K♦	Player 3	Player 3 keeps K♦	P3 +3
4	8♣	8♦	4♠	Tie (P1, P2)	P1 keeps 8♣, P2 keeps 8♦	P1 +1, P2 +1
5	2♥	10♦	10♠	Tie (P2, P3)	P2 keeps 10♦, P3 keeps 10♠	P2 +1, P3 +1
6	J♦	6♠	Q♣	Player 3	Player 3 keeps Q♣	P3 +3

Scores After 6 Rounds (Before Bonus)

Player	Round Points	Explanation
Player 1	2	Tied for highest rank in Rounds 2 and 4 (1 point each).
Player 2	6	Won Round 1 (3 points) and tied in Rounds 2, 4, and 5 (1 point each).
Player 3	7	Won Rounds 3 and 6 (3 points each) and tied in Round 5 (1 point).

Cards Collected by Each Player / Cards Collection after replacement (optional)

Player	Original Collection	Collection after replacement
Player 1	A♠, 8♣	A♠, 7♦
Player 2	K♣, A♦, 8♦, 10♦	6♠, A♦, 8♦, 10♦
Player 3	K♦, 10♠, Q♣	K♦, 8♠, Q♣

Bonus 1: Longest Consecutive Rank Sequence

Each player sorts their collected cards by rank and identifies their longest consecutive sequence.

- **Player 1:** ranks {A, 7} → longest consecutive sequence length = 1
- **Player 2:** ranks {A, 10, 8, 6} → longest consecutive sequence length = 1
- **Player 3:** ranks {K, Q, 88} → longest consecutive sequence length = 2 (Q–K)

Result: Player 3 has the longest sequence and receives **+5 bonus points**.

Bonus 2: Highest Suit Count

Each player counts how many cards they have in each suit and identifies their highest count in any single suit.

- **Player 1:** Spades = 1, Clubs = 0, Diamonds = 1, Hearts = 0 → highest suit count = 1
- **Player 2:** Diamonds = 3 (A♦, 8♦, 10♦), Clubs = 0, Spades = 1, Hearts = 0 → highest suit count = 3
- **Player 3:** Clubs = 1, Diamonds = 1, Spades = 1, Hearts = 0 → highest suit count = 1

Result: Player 2 has the highest suit count and receives **+5 bonus points**.

Final Scores

Player	Round Points	Bonus Points	Final Total
Player 1	2	0	**2**
Player 2	6	+5	**11**
Player 3	7	+5	**12**

Winner

Player 3 achieves the highest final score with **12 points** and is therefore declared the winner.

Challenge

This task is to produce a complete implementation of **High Card Series**, a card game in which two to five players compete over a nominated number of rounds between five and ten. In each round, every player is dealt one card from the deck, and the cards are compared by rank, with the player holding the highest-ranked card retaining the card and receiving three points, while ties for the highest rank result in each tied player retaining their card and receiving one point. After all rounds have been completed, each player is given a single optional opportunity to discard one or two collected cards and draw replacements from the deck, after which the collected cards are evaluated to determine the longest consecutive rank sequence and the highest suit count achieved in any single suit. The corresponding bonus points are applied, final scores are calculated, and the player with the highest total score is declared the winner.

Note: A demonstration video on Blackboard will illustrate an example submission application running.

Required Functionality

There are **9 levels** to which the submission can be completed, attracting increasing rewards.

- **Level 1:** The number of players (between **two and five**) is selected and each player is named. A standard deck of cards is created and shuffled. In each round, every player is dealt **one card**, and the card dealt to each player is displayed in the console.
- **Level 2:** For each round, the application compares the ranks of the cards dealt and determines the round outcome. The player with the highest-ranked card retains the card and is awarded **three points**. If two or more players tie for the highest-ranked card, each tied player retains their card and is awarded **one point**. Cards dealt to non-winning players are **returned to the bottom of the deck**. Updated scores are displayed after each round.
- **Level 3:** After player names are entered (in Level 1), the number of rounds is selected (minimum **five**, maximum **ten**). The application runs for the nominated number of rounds and maintains a running score for each player.
- **Level 4:** At the end of the final round, the application displays the **full set of cards collected** by each player and a summary of points accumulated during the rounds.
- **Level 5:** The application provides a **single optional adjustment stage** before bonus points are calculated. Each player may choose to discard **one or two collected cards** and draw the same number of replacement cards from the deck. This option may be used **at most once** per player and may also be skipped.
- **Level 6:** The application calculates and applies the **Longest Consecutive Rank Sequence** bonus once only at the end of the game. Each player's collected cards are analysed to identify their longest sequence of consecutive ranks. The player with the longest sequence receives **five bonus points**; if two or more players tie, each tied player receives **two bonus points**. The sequence length and bonus awarded are displayed.
- **Level 7:** The application calculates and applies the **Highest Suit Count** bonus once only at the end of the game. Each player's collected cards are analysed to count cards per suit and identify each player's highest suit count. The player with the greatest such count receives

five bonus points; if two or more players tie, each tied player receives **two bonus points**. Suit counts and bonuses are displayed.

- **Level 8:** After all bonuses have been applied, the application displays each player's **final total score** in **descending order** and declares the winner. If two or more players share the highest final score, the game is declared a **draw**.
- **Level 9:** An option for a **computer-controlled player** is provided. If a player's name is entered as **"Computer"**, the player is controlled by the application and participates in all rounds as normal. No strategic decision-making is required; the computer follows the same rules as a human player. Any assumptions or behaviours must be clearly described in the **Design and Development** document. Additional advanced functionality or presentation enhancements may also be included and must be documented accordingly.

Deliverables

The deliverables for this assessment are as follows:

1. **"High Card Series" Application** – A Zip file containing all code developed for the application. The easiest way to create this is to Zip the contents of the **src** folder of your Java project. Each Java file should also be provided as a PDF document.

2. **Design and Development Document** – A PDF document that contains a written account of the design and development decisions made during the development of your application. You should provide evidence of your consideration of data structures used and algorithms developed, with justification for your decisions and rejection of alternatives.

3. **Testing Document** – A PDF document that describes your testing regime with details of any test classes developed. This document should also describe steps you have taken to ensure the security of your implementation.

4. **Demonstration Video** – An individual video of no more than 5 minutes duration, demonstrating your application running and a walkthrough of your code base with an associated commentary concentrating on the parts of the code for which you were responsible. You do not have to appear on video in person, but a voiceover

   commentary is mandatory. Videos should preferably be recorded using Panopto (available in the module area on Blackboard), though other packages may be used provided the video is in MP4 format. Please note that videos with a duration longer than 5 minutes will be penalised according to the following scheme.

5 minutes + 10%	no penalty
5 minutes + 10% - 19%	reduction in total mark by 5%
5 minutes + >20% - 29%	reduction in total mark by 10%
5 minutes + >30% - 39%	reduction in total mark by 15%
5 minutes + >40% - 49%	reduction in total mark by 20%
5 minutes + >=50%	maximum total mark achievable is 40%

VERY IMPORTANT: The application code and video demonstration are compulsory components. Your submission will not be complete and cannot be marked if either of these elements are missing.

Submission

All 4 deliverables must be submitted to the ***Assignment 2 Submission*** link on Blackboard by the due date. Late submissions can only be accepted through the standard EC1 process.

Feedback

Feedback on each final submission will be provided in written form under the categories presented in the marking criteria that follows this specification. All results and feedback will be returned withing 20 working days of submission in accordance with university policy.

Plagiarism Declaration

By making a submission, you will be deemed to have made the following declaration of ownership. Source: https://www.ulster.ac.uk/learningenhancement/cqe/assessment-code-ofpractice/coursework-submission.

"I declare that this is all my own work. Any material I have referred to has been accurately referenced and any contribution of Artificial Intelligence technology has been fully acknowledged. I understand the importance of academic integrity and have read and understood the University's General Regulation: Student Academic Integrity and the Academic Misconduct Procedure. I understand that I must not upload my work before, during or after submission to any unapproved plagiarism detectors or answer sharing platforms, or equivalent, and that only University-approved platforms should be used."

The University policies on plagiarism and AI are available at:

- https://www.ulster.ac.uk/student/exams/cheating-and-plagiarism
- https://www.ulster.ac.uk/learningenhancement/cqe/strategies/ai

Submission Assessment Grid

Student:					
	Poor (<40%)	Satisfactory (40-59%)	Good (60-69%)	Excellent (70%+)	Mark
Design and planning (15 marks)	Limited understanding of fundamental ADT and algorithm design concepts shown. Minimum set of operations defined Lack of discussion of ADT and algorithm design and/or poor justification for design choices made	Some understanding of fundamental ADT and algorithm design concepts shown. Reasonable approach to design but lacking detailed analysis and evidence of decision making.	Discussion of ADT and algorithm design with good justification for design choices made	Demonstrates excellent awareness of design alternatives and clear, logical justification for decisions made.	
Implementation (15 marks)	Limited knowledge of ADT and algorithm development techniques shown. Minimum set of ADT operations demonstrated. Major issues with implementation of data representation.	Some knowledge of ADT and algorithm development techniques. Adequate set of ADT operations clearly demonstrated. Some issues with implementation of data representation.	Good knowledge of ADT and algorithm development techniques. Majority of ADT operations clearly demonstrated. Minor issues with implementation of data representation.	Excellent knowledge of ADT and algorithm development techniques shown. Wide set of appropriate ADT operations clearly demonstrated. Insightful discussion of most technically challenging aspects.	
Functionality (40 marks)	**Less than L1.** A game that is not playable, with significant components missing or not working (e.g., rounds cannot be completed, scoring is incorrect, or cards are not handled correctly).	**L1.** The number of players (between two and five) is selected and each player is named. A standard deck of cards is created and shuffled. In each round, each player is dealt one card and the dealt cards are displayed. **L2.** For each round, the application correctly compares the ranks of the cards dealt and determines the round outcome. The player with the highest-ranked card is awarded three points and keeps the card. Where two or more players tie for the highest rank, each tied player is awarded one point and keeps their card. **L3**. The application correctly manages the deck across rounds, returning non-winning cards to the bottom of the deck and maintaining	**L4.** The number of rounds to be played is selected (minimum five, maximum ten). The application runs for the nominated number of rounds and displays updated scores after each round. **L5.** At the end of the final round, the application displays the full set of cards collected by each player and a summary of points accumulated during the rounds. **L6**. The application implements the optional end-of-game adjustment stage, allowing each player a single opportunity to discard one or two collected cards and draw replacements from the deck before bonus points are calculated.	**L7.** The application correctly calculates and applies the bonus scoring rules at the end of the game, including: - the Longest Consecutive Rank Sequence bonus, and - the Highest Suit Count bonus, with correct handling of ties in both cases. **L8.** An option for a computer-controlled player is provided. If a player's name is entered as "Computer", the player is controlled by the application and participates in the game according to the same rules as a human player. Any assumptions or automated behaviour are clearly explained. **L9.** Any other advanced functionality or presentation that enhances gameplay or usability, such as	

		a running score and card collection for each player throughout the game.		improved console formatting, robust input validation, detailed end-of-game statistics, or saving and loading game results. Any additional functionality must be clearly explained and justified.	
Testing (20 marks)	No evidence of testing or no evidence that testing has been effective.	Details of basic testing are present but no evidence of a structured approach. Limited documentation of the testing process.	Some test classes have been provided, but with less than full coverage. Incomplete testing documentation.	All main classes have test classes that give proper attention to the range of potential input values including edge cases. A comprehensive write-up that fully documents the testing process.	
Video (10 marks)	Video lacking in some element such as voiceover, code walkthrough, or game demonstration	Video is present but is short or lacking in detail in the code walkthrough and/or game demonstration.	Video demonstrates functionality and discusses the main code elements – but does not provide any insight beyond what can be seen by running the software.	Video provides additional insight into the development process. Excellent demonstration of working functionality.	
		Initial Mark:	**Deductions:**	**Total Mark:**	